A Bibliometric analysis of research papers using Graph Neural Networks and Large Language Models

Project report submitted to the Amrita Vishwa Vidyapeetham in partial fulfilment of the requirement for the Degree of

BACHELOR of TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **"Bibliometric analysis of research papers using Graph Neural Networks"** submitted by **Janvi P(AM.EN.U4CSE20139), Kavya Krishnan(AM.EN.U4CSE20139), Poonam C Sanil(AM.EN.U4CSE20151) and Sthuthi Rakesh Raja(AM.EN.U4CSE20166))**, in partial fulfillment of the requirements for the award of Degree of Bachelor of Technology in Computer Science and Engineering from Amrita Vishwa Vidyapeetham, is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Amritapuri during Semester 8 of the academic year 2023-2024.

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DECLARATION

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Place : Amritapuri

Date : 15 May 2024

Signature of the student

Signature of the Project Guide

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Furthermore, we recognize and appreciate the significant contributions of all team members to this project. Each member's dedication, hard work, and collaborative spirit have played a pivotal role in achieving the success and completion of this endeavor.

We extend our sincere thanks to our reviewer for their constructive feedback and to our coordinators for their valuable insights and guidance, which have immensely contributed to the project's direction and success. Additionally, we express our gratitude to the Department of School of Computing for their continuous support and encouragement, which has been instrumental in our journey.

Abstract

In the dynamic landscape of academic research, the integration of sophisticated models within user-friendly interfaces plays a pivotal role in enhancing user experience and facilitating efficient information retrieval. This project endeavours a novel approach aimed at seamlessly integrating advanced models into a user-friendly interface, with a primary focus on enhancing the user experience by integrating a chatbot - Samvidh which lets the user ask queries based on any particular research paper and also engage in general conversation. In the realm of academic research, the accessibility and usability of advanced models are crucial for facilitating efficient information retrieval and insightful analysis. Central to this endeavor is the development of an intelligent web interface designed to streamline the retrieval of relevant papers, facilitate intuitive visualization of citation networks, and present classification results in a clear and comprehensible manner. Leveraging the capabilities of Graph Neural Networks (GNNs) and Large Language Models (LLMs), our interface offers researchers and practitioners powerful tools for efficient information extraction and analysis. A key contribution of this work is the integration of Samvidh, a chatbot designed to enhance the user experience by providing natural language query capabilities, seamless paper retrieval functionalities, and interactive communication channels. Samvidh serves as an intelligent assistant, allowing users to interact with the system in a conversational manner, thereby reducing the cognitive load associated with traditional search interfaces. Through collaborative efforts, GNNs, LLMs, and Samvidh work synergistically to revolutionize the landscape of advanced paper analytics techniques and user-centric interfaces. By

empowering users with powerful tools and intuitive platforms, this research aims to facilitate seamless navigation and extraction of valuable insights from the vast expanse of scholarly literature.

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Chapter 1

Introduction

The rapid expansion of the research landscape presents an increasingly daunting challenge when it comes to effectively navigating and extracting relevant information from the ever-growing pool of scholarly literature. This exponential growth not only highlights the richness of available knowledge but also underscores the need for advanced tools and methodologies to make sense of this vast amount of data. In response to this challenge, our project integrates a responsive chatbot, Samvidh, designed to provide quick access to information within this vast scholarly landscape. By leveraging natural language processing capabilities, the chatbot offers users an intuitive interface to swiftly address basic queries related to research papers, serving as a key component in overcoming the complexity of navigating the burgeoning research landscape.

Among the cutting-edge technologies that have emerged to tackle this challenge, Graph Neural Networks (GNNs) stand out as particularly promising. GNNs have the unique ability to uncover hidden patterns and connections within graph data, offering researchers a deeper understanding of the underlying structure of academic literature. By leveraging the inherent rela-

tionships and dependencies encoded in graph-structured data, GNNs enable more nuanced and comprehensive analysis compared to traditional search methods.

GNNs offer a robust methodology for identifying intricate relationships and dependencies in graph-structured data. Their ability to capture complex interactions makes them well-suited for analyzing the multidimensional and interconnected nature of research datasets. By incorporating GNNs into our analytical framework, we aim to enhance the accuracy, depth, and scope of our bibliometric analyses, ultimately facilitating more informed decision-making and knowledge discovery within the academic community.

Our objective is to meticulously curate a diverse bibliometric dataset that encompasses various academic domains, serving as the foundation for our analytical pursuits. Through the utilization of graph neural networks (GNNs), we plan to deploy a range of sophisticated techniques including node classification, link prediction, and community detection to conduct comprehensive graph analytics on this dataset.

Concurrently, we are dedicated to developing a user interface that is intuitive and user-friendly. This platform will leverage insights derived from GNNs to offer visualizations depicting relationships between research papers, personalized recommendations, detailed hover information, and a dynamic three-dimensional view of the dataset. These features aim to enhance user engagement significantly.

Our methodology emphasizes continuous evaluation and iterative improvement. We actively solicit feedback to assess the effectiveness of our approach, ensuring that our system not only meets user requirements but also evolves to enhance usability and functionality, thereby contributing significantly to the field of bibliometric analysis and research exploration.

Chapter 2

Problem Definition

Navigating the expansive realm of academic literature poses significant challenges, as traditional search methodologies often fail to yield nuanced results tailored to individual needs. This study endeavors to rectify this issue by harnessing the power of Graph Neural Networks (GNNs) to conduct thorough bibliometric analyses. However, notable obstacles persist, including the incapacity of existing search paradigms to discern semantic relationships effectively and the absence of comprehensive tools for holistic research landscape visualization.

Therefore, the primary problem statement lies in developing a sophisticated solution that seamlessly integrates Graph Neural Networks with an intuitive user interface, thereby unraveling latent connections between scholarly works. This endeavor holds promise for enhancing research exploration and facilitating more efficient discovery processes.

Chapter 3

Related Works

The related works encapsulates an examination of existing literature and practical implementations directly relevant to the scope of this study. This segment is structured into two primary categories: application references and paper references.

3.1 Application References

Application References delves into real-world applications, tools, or systems that have a direct connection to the subject matter of this project. These references serve as practical demonstrations of concepts or technologies discussed within the context of the study.

Application References			
Index	Application Name	Function	Difference
1	Research Rabbit	AI-powered platform designed to help researchers discover relevant academic papers, enabling efficient literature search and exploration across various disciplines.	Provides a graph visualization with hover-on-node details, subpages for selected papers offering additional features like full PDF access and paper-specific chatbots.
2	Connected Papers	A web-based tool that helps researchers visualize connections between academic papers, facilitating exploration of related work and identifying key research trends and influences.	Offers a broader range of features beyond visualization, including subpages with detailed paper information and additional tools like paper-specific chatbots. Connected Papers primarily focuses on visualizing connections between academic papers without the extensive supplementary features.

Application References			
Index	Application Name	Function	Difference
3	Litmaps	A platform that creates interactive visual maps of scientific literature, aiding researchers in exploring and understanding complex research landscapes.	Incorporates a graph visualization with interactive node details and subpages for selected papers, providing extra utilities such as full PDF access and paper-specific chatbots. While Litmaps focuses on creating interactive visual maps of scientific literature, it may not offer comparable supplementary features.
4	Google Scholar	A freely accessible web search engine that indexes scholarly literature across various disciplines. It provides a vast database of academic articles, theses, books, conference papers, and patents, aiding researchers in finding relevant sources for their work.	Provides a curated platform with enhanced user interaction features such as graph visualizations, paper-specific chatbots. While Google Scholar offers a vast database of scholarly literature, it may lack the personalized and interactive features.

3.2 Paper References

Paper References encapsulates a collection of scholarly works, research papers, and academic publications that contribute significantly to the theoretical framework and knowledge base of this study. These references are carefully curated to encompass a broad spectrum of perspectives, methodologies, and findings relevant to the project objectives.

1. Link Prediction Based on Graph Neural Networks

The paper[7] introduces a novel methodology for link prediction in network-structured data, deviating from conventional approaches by emphasizing learning heuristics directly from the network rather than relying on predefined ones. This shift is motivated by the observation that traditional link prediction heuristics often make strong assumptions about when two nodes are likely to link, which may not hold in diverse network contexts, thus limiting their effectiveness. To address this, the paper proposes a Graph Neural Network (GNN)-based algorithm for learning heuristics, which demonstrates superior performance across a range of network problems.

However, while the proposed approach shows promise, it also exhibits certain limitations that warrant consideration. One such limitation is its potential lack of generalizability across different types of networks. It may require extensive fine-tuning to achieve optimal performance in diverse scenarios. Moreover, implementing the GNN-based algorithm for heuristic learning can be computationally expensive, particularly for large-scale networks, posing challenges to scalability.

Transductive learning struggles in large graphs, supervised graph analysis faces challenges with unlabeled data, and GNNs diversify node classification applications. The identified open problems revolve around scalable graph

algorithms, improved inductive learning, and effective task-transformation strategiesIn light of these concerns, future research directions could focus on enhancing the scalability of heuristic learning algorithms, especially by exploring methods to efficiently process large-scale networks with millions of nodes and edges. Additionally, there is a need to address the robustness of the proposed approach to variations in network structure and noise. This entails developing strategies to handle noisy or incomplete data and adapt the learned heuristics accordingly, thereby ensuring the reliability and applicability of the methodology in real-world network settings.

2. Graph Neural Networks: A bibliometrics overview(2023)

[8] addresses the need for a comprehensive understanding of the research landscape in Graph Neural Networks (GNNs) since their inception in 2004. Through a Scopus-based bibliometric analysis, the study aims to quantitatively and qualitatively evaluate trends, subjects,3 authors, institutions, sources, citations, and emerging topics in GNN research, highlighting key challenges and future directions in the field.

In-depth bibliometric analysis surveys Graph Neural Networks (GNNs) research from 2004, examining trends in subjects, authors, institutions, publications, and research topics. Limitations encompass reliance on Scopus, potential subjectivity in qualitative assessments, static timeframe since 2004, oversimplified task categorization, and institutional citation bias. The identified open challenges include scalability, generalization, and over-smoothing in models, as well as the need for improved explainability, ethical considerations, and adaptability. Additionally, benchmarks, transfer learning, and robustness against adversarial attacks pose crucial areas for exploration.

3. Chatbots in Knowledge-Intensive Contexts: Comparing Intent and LLM-Based Systems (2024)

The paper addresses [9] the limitations of traditional intent-based cognitive assistants in handling the diverse nature of natural language during knowledge-intensive tasks. While recent advances in natural language processing, such as large language models like GPT-4, Llama2, and Gemini, offer potential for more flexible and human-like interactions, concerns about unforeseen consequences in accuracy-critical contexts arise. A user study comparing LLM-based and intent-based systems revealed superior performance of LLM-based approaches, prompting further investigation into transitioning NLP techniques in knowledge-intensive contexts.

Furthermore, the paper acknowledges the communication and education potential of LLMs, like ChatGPT, for their ability to facilitate language-like communication and provide educational support. However, it emphasizes the ongoing necessity for research to understand LLMs' social implications and broader societal impact, while also raising awareness about ethical concerns, including privacy, data rights, and the potential for misuse.

In response to these challenges, the paper calls for responsible leadership and collaborative efforts among leaders to ensure the responsible use of LLMs, prioritizing humanity's well-being. It recognizes the potential for misuse of LLMs and suggests the need for regulatory guidance to balance innovation with ethical considerations. Additionally, the paper highlights the importance of measures to ensure beneficial LLM use, along with the need for educational initiatives to teach LLM fundamentals and ethics. Nonetheless, open questions remain regarding how governments can assess and mitigate potential negative impacts of LLMs on society.

4. Enhancing Efficiency in Parallel Louvain Algorithm for Community Detection(2023)

This paper discusses [10], the significance of community detection in network analysis, particularly in identifying groups and patterns within networks, which is crucial for various applications. With the growing size of networks, there is a pressing need for fast algorithms to analyze them efficiently. One such algorithm discussed is the modularity-based greedy Louvain algorithm, known for effectively dividing networks into disconnected communities over several iterations. Despite its ability to establish high-quality communities, it can be slower compared to label-propagation-based techniques, which sacrifice quality for speed. However, researchers have proposed methods for parallelizing and improving the Louvain algorithm to enhance its performance and scalability.

In their investigation, the authors implement single-threaded and multi-threaded versions of the static Louvain algorithm, examining its specifics, optimizing parameters, and evaluating performance. They find that adjusting the tolerance between each pass and employing an asynchronous version of the algorithm yields the best results, despite limitations in shared-memory parallelism. The paper suggests potential strategies, such as breaking the graph into manageable chunks for independent execution, to overcome scalability challenges associated with the algorithm.

Moving beyond community detection, the paper discusses the broader context of parallel algorithms for graph analysis, highlighting the increasing use of graphs to represent interconnected data from real-world sources. It emphasizes the importance of parallelism in handling massive and dynamic graphs, which require efficient algorithms for analysis and updating. The authors cite examples of research into dynamic graph analytics, such as centrality

score calculation, biconnected component maintenance, and shortest path computation, underscoring the significance of parallel algorithms in various graph-related tasks.

5. Graph Convolutional networks: analysis, improvements and results(2021)

The work [11] enriches topological information, redesigns networks with dense layers, and introduces minor improvements. Benchmark results show competitive and state-of-the-art performance, with lower computational costs compared to alternatives.

Enhanced graph convolutional networks improve semi-supervised learning accuracy with structural modifications and parameter optimization, outperforming competitors in efficiency. Limited exploration of 2D or 3D weighting schemes; potential for further optimization and generalization. The paper identifies two crucial challenges: dynamic graph adaptation for evolving networks and the creation of interpretable graph representations, critical for model adaptability and interpretability.

The study presents enhancements of GCN and ConfGCN for the task of semi-supervised learning with graph convolutions. It focuses on four main modifications: parameter configuration, integration of additional structural information into adjacency matrices, convex optimization of activation functions, and combination of base networks with dense layers. These improvements aim to boost accuracy, contrasting with baseline networks where the addition of new layers tends to decrease accuracy. While Graph Convolutional Layers currently employ 1D convolutions, potential exploration of 2D or 3D weighting schemes exists. GCN initially proposed a layer-wise propagation rule for SSL, while ConfGCN was subsequently introduced as a

network estimating label scores with label confidences. Six different network configurations are proposed and validated on four benchmark datasets, with parameter optimization facilitated through grid search. This approach yields high accuracy and low execution times across all datasets.

Chapter 4

Requirements

This outlines key software elements vital for the project's functionality. These include tools for dataset management, development environments, graph neural networks, large language models, user interface development, data visualization, and database management. These requirements form the core framework supporting the project's architecture and functionalities.

4.1 Software

The project's design is structured around several software components, each outlined comprehensively in the specifications provided below. These specifications detail the essential software elements that form the foundation of the project architecture and functionality.

- Dataset : Cora citations[12] using Planetoid and by loading Cora content dataset.
- Integrated Development Environments: IDEs like Google Colab and Visual Studio Code (VSCode), offer comprehensive tools for coding, debugging, version control, and collaboration.

- GNN frameworks: Specialized libraries and tools like PyTorch Geometric etc designed for working with graph-structured data and implementing various graph neural network architectures.

- Large Language Model (LLM) Framework: Examples include Langchain, openAI, word2vec, streamlit and PY2PDF.

- UI Development Tools: This comprehensive set of tools ensured a seamless and engaging user experience, with Flask handling data processing and rendering, HTML providing the structural foundation, CSS enhancing visual aesthetics, and JS adding interactivity to the interface.

- Visualization Tools: For data visualization, we used tools including Torch for advanced data processing, Plotly for interactive and customizable plotting, Pandas for efficient data manipulation and analysis, and NetworkX for creating and visualizing graph networks.

- Database Management System(DBMS): Python utilizes its default RDBMS, SQLite for storing user data.

Chapter 5

Proposed Methodology

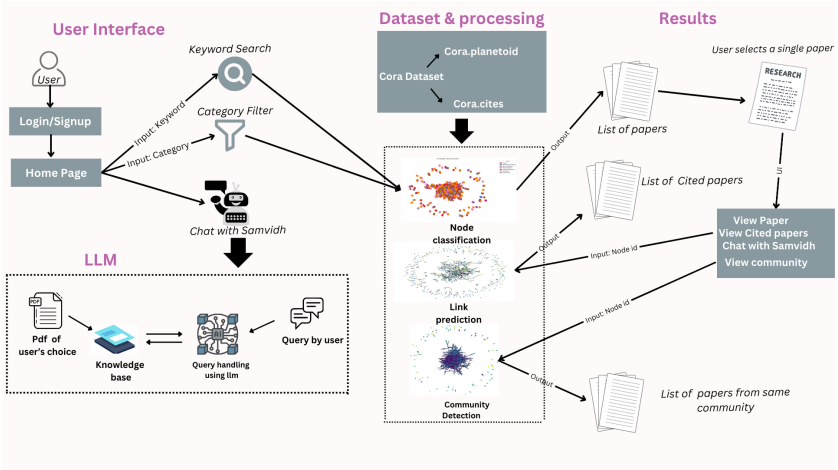


Figure 5.1: Workflow

To address the challenge of efficient research paper discovery, this project proposes a novel application that leverages the Cora dataset. The core functionality revolves around an interactive 3D visualization where research papers are represented as nodes and their connections depicted by links. Color-coding facilitates the immediate identification of thematic categories, allowing users to grasp the research landscape at a glance. Furthermore, a robust keyword search function empowers users to locate specific papers based on title, author, or keywords within the content. Indexing techniques

will ensure rapid retrieval of relevant results. To further refine exploration, a granular category filtering mechanism is implemented, enabling users to target specific research domains within the Cora dataset. This targeted approach promotes a more efficient discovery process. Recognizing the value of inter-paper relationships, the application facilitate exploration of connections between research papers. Additionally, a mechanism is developed to assess the community relevance of individual papers, potentially incorporating citation data or other metrics to highlight highly influential works within a specific domain. Finally, to enhance user experience, a user-friendly chatbot named Samvidh is integrated within the application. Samvidh will utilize natural language processing techniques to address basic queries related to specific research papers within the collection. This multifaceted approach, combining visualization techniques, intelligent search algorithms, and user-friendly interfaces, aims to create a comprehensive research paper discovery platform that empowers researchers and scholars to navigate the ever-growing body of research literature with greater ease and efficiency.

5.1 User Interface

5.1.1 Registration and Login

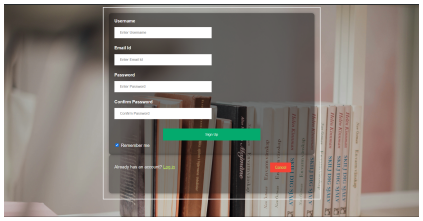


Figure 5.2: New User Sign-Up Page

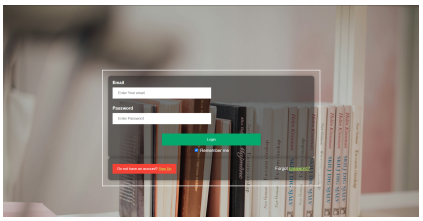


Figure 5.3: Login Page

A new user can sign-up to the website using register page (Figure 5.2) and the user credentials will be stored in the database. After signing up the user

will be authenticated in the login page (Figure 5.3) only the verified user will be able to access the website.

5.1.2 Home

The Home page (Figure 5.4)of the UI is divided into three main parts, the side bar, the map bar and the search bar.

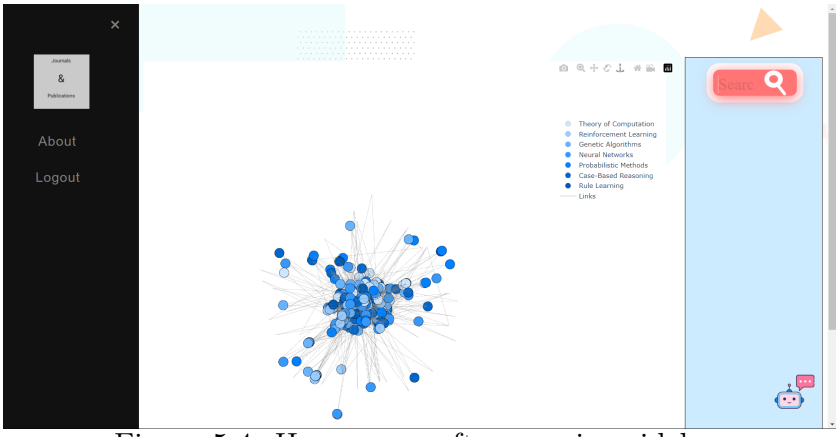


Figure 5.4: Home page after opening sidebar and searching for paper

1. The side bar (figure 5.4) has two buttons, an About button and a profile button.
   (a) The About button opens up the About page, where it contains the summary of the website and authors.
   (b) This button enables users to securely log out of their accounts.
2. The map bar is a 3D visualization of the classified CORA dataset. By hovering over each node user can see the paper details of the hovered node. The classifications of each paper are given in the top right corner from which users can find papers that are of desired category or class.
3. The search bar is integrated with a Node Classification GNN model to enable keyword search and find papers with the searched keyword.

The paper list shown can redirect you to the paper details in the Paper Details page.

5.1.3 Key Paper Insights



Figure 5.5: Paper details of the selected paper

The Paper Details page (Figure 5.5) of the UI is where the users can view papers and read them. The three main components of the Paper Details page are The paper, Cited papers, Related Papers and Relevance score in the community. The desired user can view and download the research paper and ask our chatbot, Samvidh about the selected paper.

1. The paper can be viewed and downloaded by the users.

2. The related papers component shows the papers that belong to the same community of the viewed paper. This is implemented using Community Detection.

3. The Cited Papers list the papers cited by the viewed paper. This is implemented using a Link Prediction model.

4. The chatbot, Samvidh helps the user with understanding and learning more about the paper.

5.2 Keyword driven paper-search

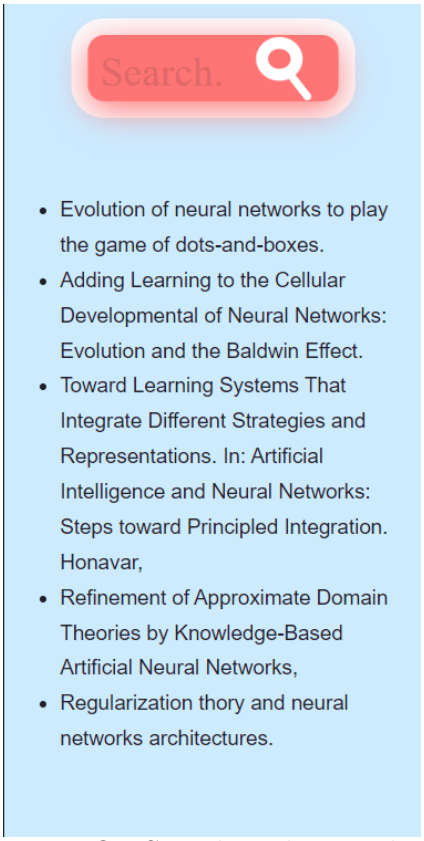


Figure 5.6: On Searching keyword neural

In the keyword search functionality, node classification enriches the search process by providing contextually relevant results. Through a Graph Convolutional Network (GCN) model trained on the Cora dataset, papers are categorized into predefined classes based on their content and citation net-

work structure. When a user enters a keyword, the system filters through paper titles, leveraging both keyword matches and the assigned categories from node classification. This integration ensures that search results not only include papers containing the specified keyword but also consider thematic relevance, aligning with the broader categories of interest to the user. As a result, users benefit from a more nuanced and contextually informed search experience, optimizing the exploration of scholarly resources.

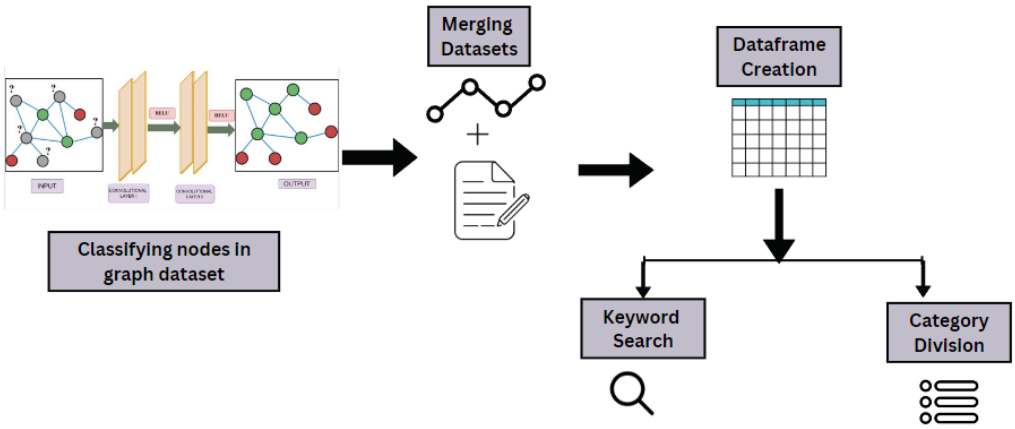


Figure 5.7: Node Classification Model

The Cora dataset, consisting of interconnected academic papers and their corresponding categories, serves as a rich resource for understanding citation relationships and topical classifications within scholarly literature. By merging the graph dataset, which encodes citation connections, with the text dataset containing paper titles and categories, a unified input is formed for analysis. Employing a Graph Convolutional Network(GCN) architecture (Figure 5.7), specifically tailored for graph-structured data, enhances our ability to discern intricate relational patterns among papers. With two con-

volutional layers and Rectified Linear Unit(ReLU) activation functions, the GCN efficiently captures and processes information, facilitating the extraction of nuanced insights from the interconnected web of academic citations.

```
Final Evaluation - Loss: 0.0102, Test Accuracy: 0.8110, Num Features: 7
```

Figure 5.8: Node Classification Model Evaluation

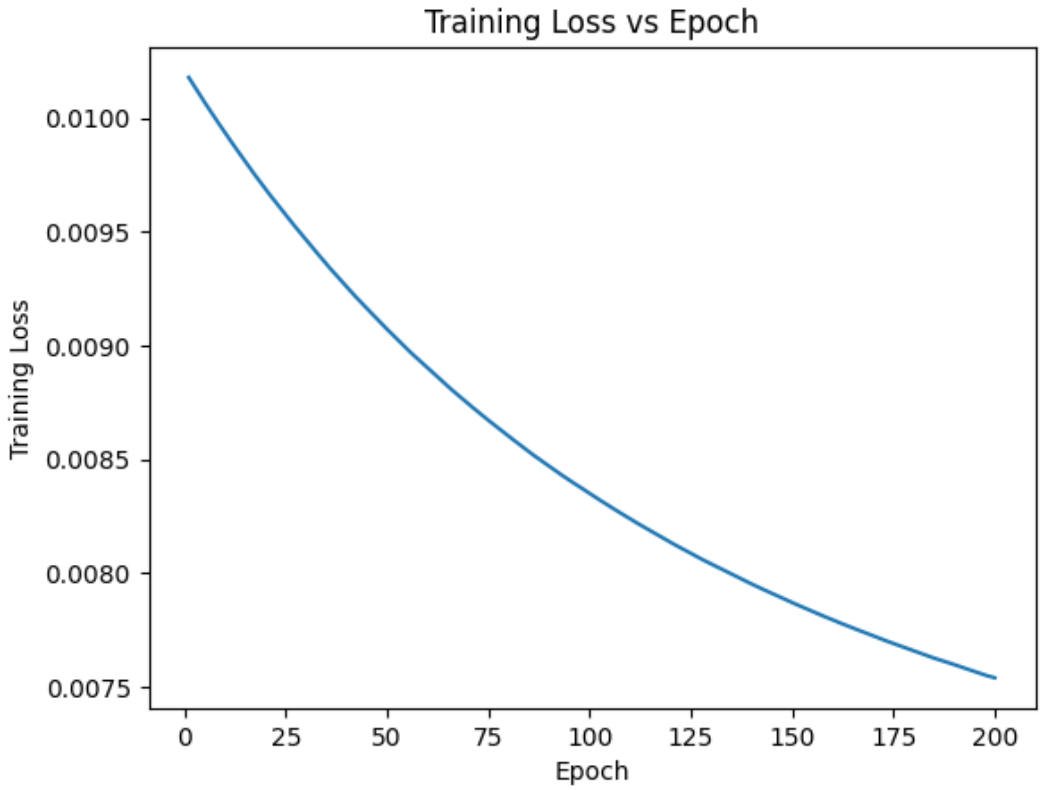


Figure 5.9: Epoch v/s Loss Graph

5.3 Implement category-specific filters on papers

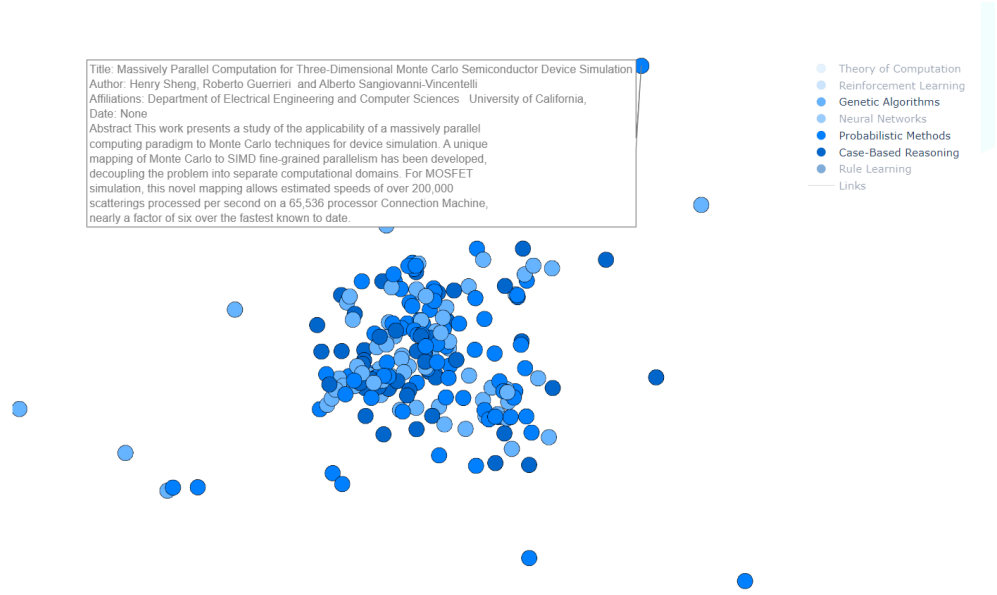


Figure 5.10: Selected category nodes and hovered node showing paper details

In order to enhance the usability and effectiveness of our dataset visualization tool, we have incorporated category-specific filters, allowing users to explore data within distinct domains of interest5.10. The dataset itself is structured around seven fundamental categories, each representing a significant field of study within our research domain:

1. Reinforcement Learning
2. Theory of Computation
3. Genetic Algorithm
4. Neural Networks

5. Probabilistic Methods

6. Case-based Reasoning

7. Rule Learning

These categories span a diverse range of topics, from machine learning methodologies like Neural Networks and Reinforcement Learning to computational theories, genetic algorithms, probabilistic approaches, case-based reasoning, and rule learning paradigms.The implementation of category-specific filters within our visualization tool allows users to selectively view papers that align with their specific areas of interest. For instance, upon selecting the "Neural Networks" category, the visualization dynamically adjusts to display nodes exclusively related to papers on Neural Networks.

Underlying this functionality is our dataset's classification using advanced techniques such as Graph Convolutional Networks (GCN) (Figure 5.7). Through node classification, each paper is categorized into one of these seven domains, facilitating seamless filtering and visualization based on user preferences. By enabling users to focus their exploration within targeted categories, our platform empowers researchers to extract meaningful insights, identify trends, and conduct in-depth analyses within their chosen domains. This feature not only enhances the user experience but also promotes efficient data exploration and knowledge discovery tailored to individual research objectives.

5.4 Personalized Paper Recommendation

Following keyword search , the application will dynamically generate a curated list of research papers (Figure 5.11) that align with the user's search criteria. This list will be tailored based on the specific keywords entered or

the chosen category filter. Users then have the autonomy to select any paper from the presented list and delve deeper by exploring the cited references within that particular paper. This functionality provides valuable context and facilitates the identification of related research that builds upon the chosen paper.

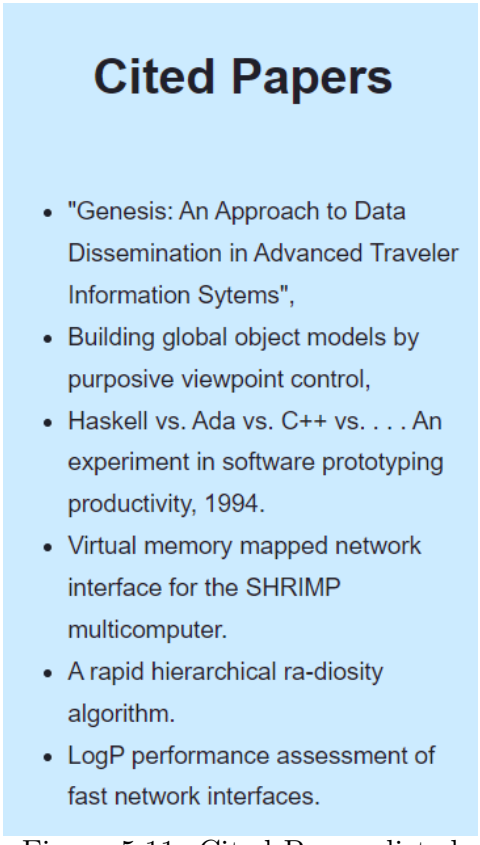


Figure 5.11: Cited Papers listed

The link prediction model (Figure 5.12) utilizes a Graph Convolutional Network (GCN) to predict relationships in a citation graph. It processes node features through two graph convolution layers, utilizing Rectified Linear Unit (ReLU) activation functions to capture the graph structure. The model is trained to predict links between nodes based on learned representations and achieves this through supervised training with cross-entropy loss

(Equation 5.1). The architecture diagram is given below:

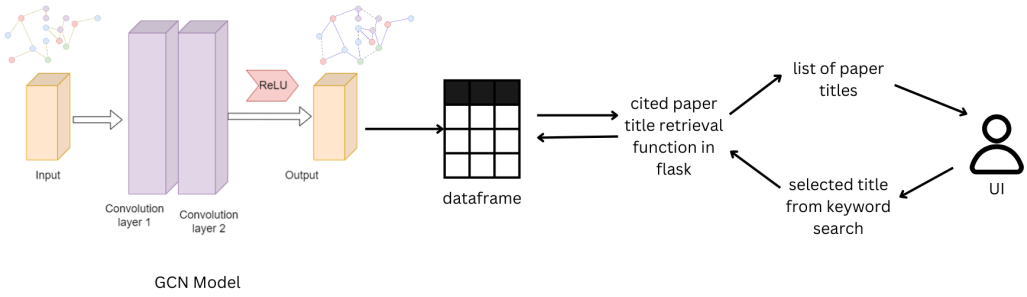


Figure 5.12: Link prediction Model Architecture

The cross-entropy loss used in this link prediction model is computed using the formula:

$$\text{Cross-Entropy Loss} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \tag{5.1}$$

where:

- y is the predicted probability distribution,
- t is the target probability distribution (one-hot encoded labels),
- N is the number of classes (number of unique labels),
- ti is the true label for class i
- yi is the predicted probability of class i

5.5 Community based paper discovery

When searching a keyword in the application, a list of papers containing the keyword is displayed. Upon selecting a paper, users can access its details including the abstract, view PDF option, cited papers and chatbot features. Additionally, more papers from the same community are presented, as shown in (Figure 5.13), facilitating the identification of topographic-based clusters within the academic network. This feature reveals thematic or structural groupings among research topics or areas, as illustrated in the accompanying figure. Our platform integrates community detection using the Louvain algorithm to dynamically display related papers in the UI, promoting interdisciplinary exploration and enhancing user experience with a cohesive view of research clusters.



Figure 5.13: Related Papers listed

The proposed community detection method integrates a node classification model using Graph Convolutional Networks(GCN) to derive meaningful

node representations from the graph structure.This model consists of two graph convolution layers with ReLU activation functions to capture structural features.After training the GCN model, the learned node embeddings are extracted and used as features for Louvain community detection on the original graph.These representations are then fed into the Louvain algorithm, optimizing graph modularity by assigning nodes to communities based on representation similarity and analyzing node influence within communities. The architecture along with model and algirthm is given in Figure 5.14.

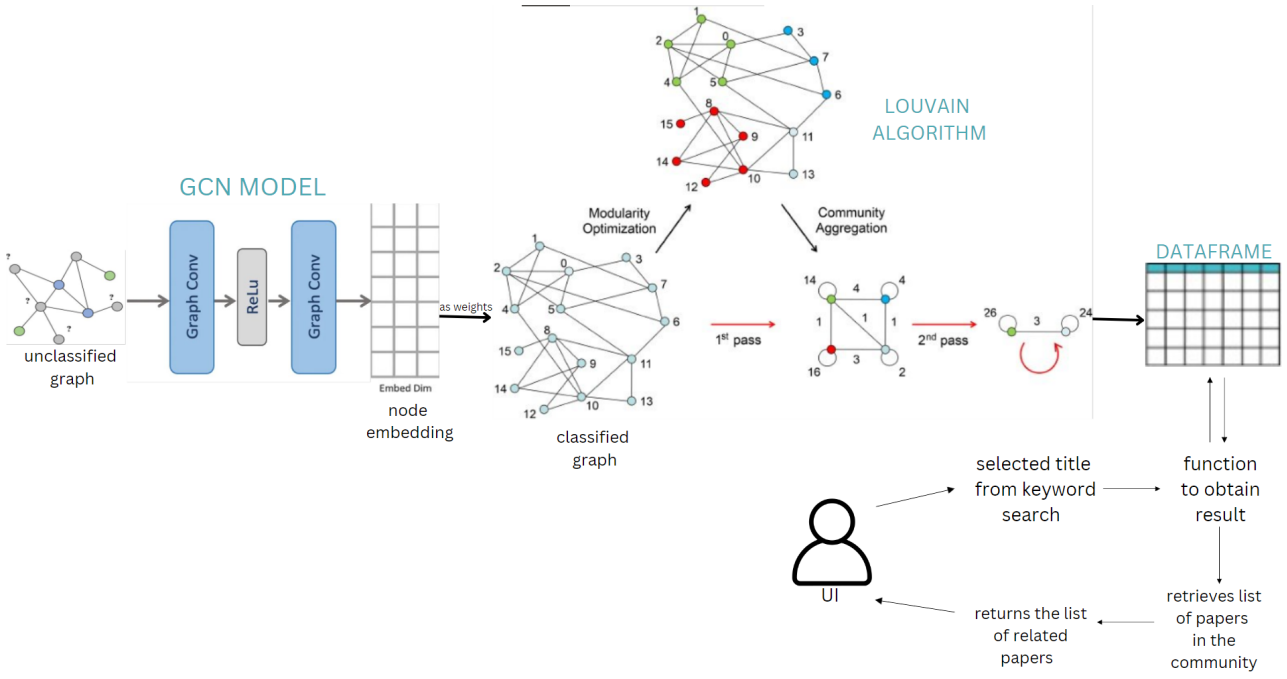


Figure 5.14: Community Detection Architecture

The model utilizes the Cross Entropy Loss (Equation 5.2), which internally applies the softmax function to the model's output.

$$\text{Cross-Entropy Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C} y_{ij}\log(\hat{y}_{ij}) \tag{5.2}$$

where:

- N is the number of samples,
- C is the number of classes,
- y_{ij} is a binary indicator(0 or 1) whether sample i belongs to class j,
- $\hat{y}_{ij}$ is the predicted probability of sample i belonging to class j, produced by model's softmax function.

5.6 Smart Virtual Assistant Bot(Samvidh)

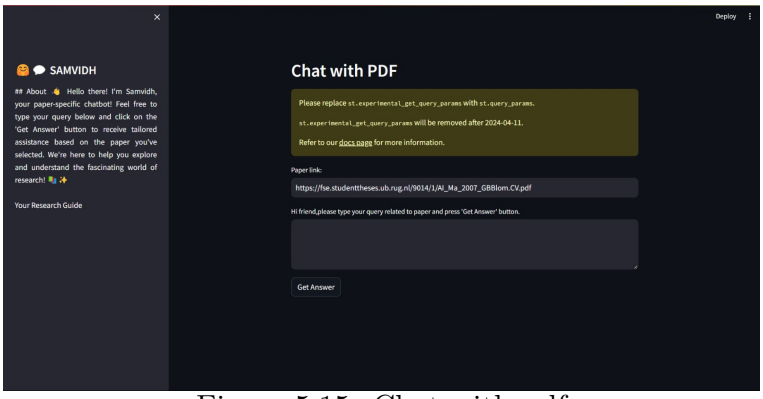


Figure 5.15: Chat with pdf

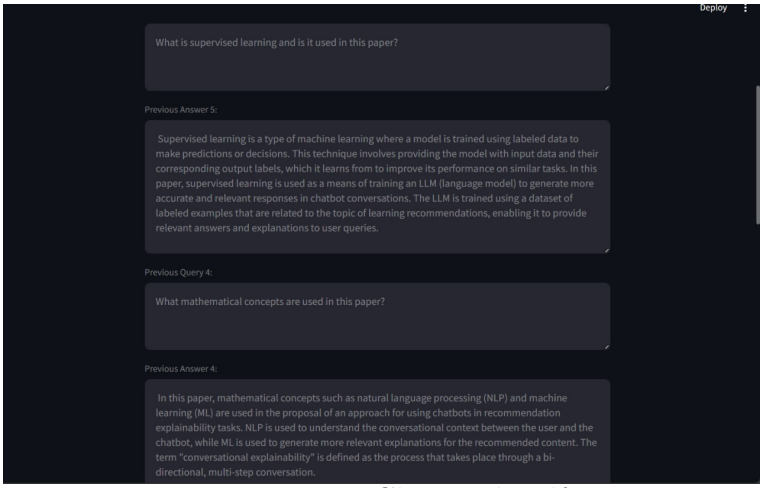


Figure 5.16: Chat with pdf

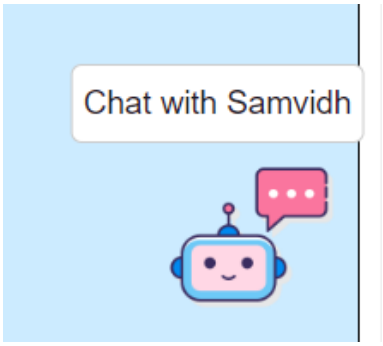


Figure 5.17: Samvidh bot

In the integrated UI, users initiate interactions with Samvidh by selecting a research paper within the interface. If users wish to delve deeper into the selected paper's content or related topics, they can engage with Samvidh (Figure 6.10). The system then processes the chosen paper, extracting its text and segmenting it into manageable chunks. These chunks undergo transformation into Word2Vec embeddings, capturing semantic nuances. Subsequently, users can pose questions about the paper, leveraging these embeddings to identify the most pertinent sections of the document (Figure **??**). Additionally, Samvidh incorporates a Large Language Model (LLM), enhancing its capability to understand and respond to user queries in a contextually relevant manner. This process enables users to engage in natural language queries, fostering an intuitive exploration and understanding of research papers facilitated by Samvidh.

In this case, the function 'getprompt' constructs a prompt that includes the user's query, along with the top-k similar documents retrieved from the PDF, providing context for the language model to generate a relevant response. This structured prompt helps guide the language model in understanding the user's query and producing coherent responses tailored to the specific context of the PDF content.

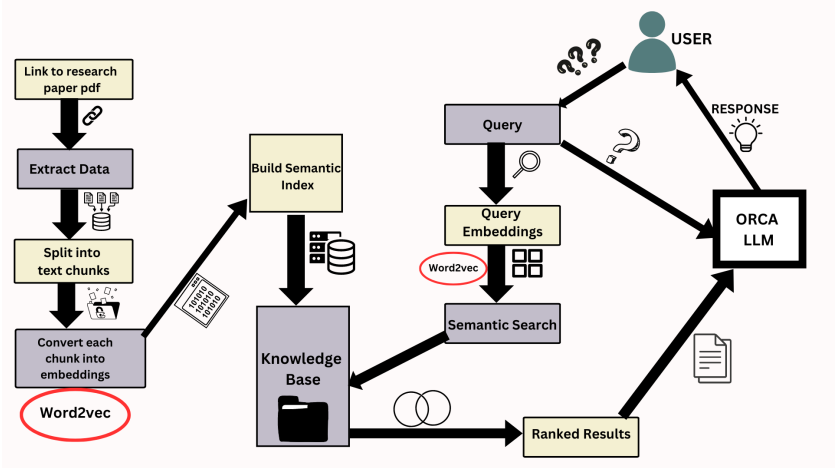


Figure 5.18: Work flow of Samvidh (Chat with Pdf)

The website's built-in AI chatbot designed to enhance your browsing experience. Users can engage in educational dialogue, seeking clarification on research papers or exploring various concepts. With Samvidh's extensive knowledge base, users have a trusted companion to answer any query, fostering a seamless journey through learning and discovery..

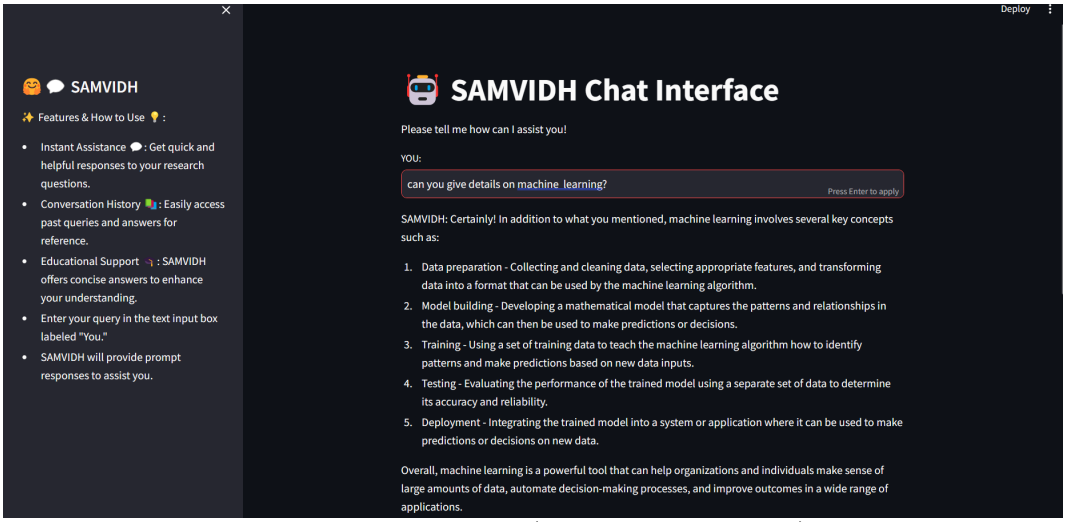


Figure 5.19: Samvidh(General Assistance)

The workflow (Figure 5.18) of interacting with Samvidh, the AI chatbot, unfolds seamlessly through a series of orchestrated steps. Users initiate

the conversation by entering their queries, prompting Samvidh to process each input methodically. Leveraging a pre-trained language model, Samvidh meticulously analyzes the user's query alongside the conversation's historical context, enabling it to generate articulate responses. The exchange continues iteratively, with Samvidh dynamically updating the conversation history to maintain contextual relevance. Users may conclude the interaction at their discretion, signaling Samvidh to gracefully bid farewell. This iterative dialogue mechanism ensures a responsive and engaging conversational experience, facilitating effective communication and knowledge dissemination.

Ssmvidh's operational framework, prompt engineering serves as a pivotal mechanism to guide the conversation flow and content generation. Through the 'getprompt' function, Samvidh meticulously crafts structured prompts that encapsulate both user queries and the historical context of the conversation. By integrating historical context into prompts, Samvidh ensures that subsequent responses remain coherent and contextually grounded. This strategic use of prompt engineering fosters a seamless and natural conversational experience, enhancing user engagement and satisfaction with Samvidh's interactions.

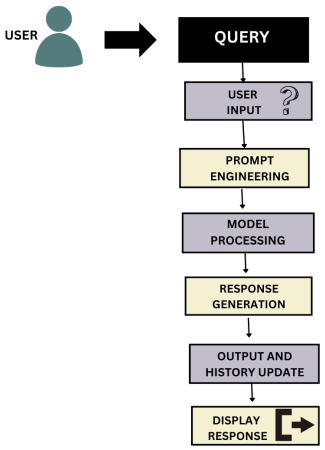


Figure 5.20: Work flow of Samvidh(General Assistance)

5.7 Accessing Research Papers



Figure 5.21: View Paper Button

When a keyword is searched in the application, it displays papers with the keyword. Upon selecting a paper, users access details like abstract, chatbot features, cited papers, and community papers. The View PDF(Figure 5.21) option redirects to the paper's PDF, mapped via a dataframe to extract title and authors for an API-based PDF extraction function. This function utilizes SerpAPI, which uses an API key to fetch Google search results in JSON format, sifting through 'organic results' for links containing 'type=pdf' or ending with '.pdf'. Matching links are added to the dataframe, accessible on clicking 'View PDF' in the UI. Working provided in Figure 5.22.

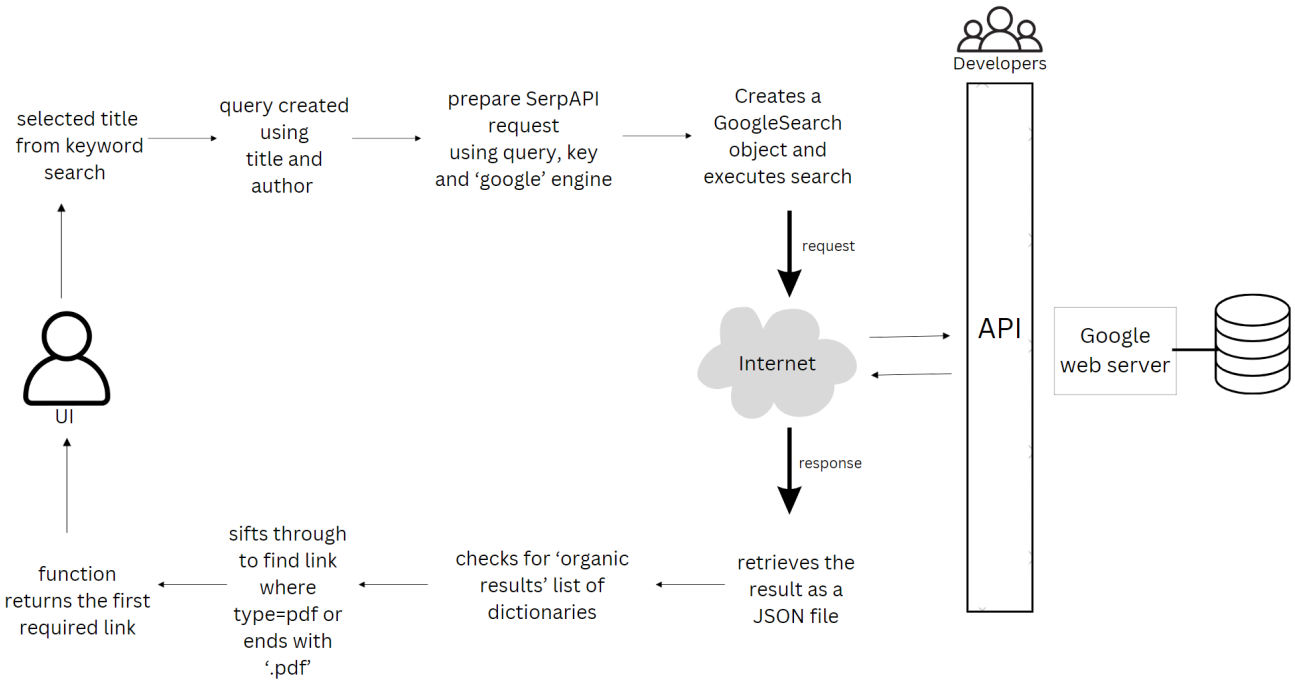


Figure 5.22: Working of PDF retrieval

5.8 Algorithms

In this project, algorithms are utilized to enhance data exploration and analysis, to uncover valuable insights and patterns within the research domain. This computational tool contribute significantly to the efficiency and accuracy of our data-driven approaches, enabling us to derive meaningful conclusions and make informed decisions based on the analyzed data.

5.8.1 Louvain Algorithm

After obtaining node representations from the Graph Convolutional Network (GCN) in the node classification model, the Louvain algorithm (Algorithm 1) is applied to identify communities within the graph. This algorithm, known for its efficiency, is utilized for community detection in the context of node classification. It optimizes modularity (Equation 5.3), a measure that evaluates the quality of community divisions in a network. The algorithm iteratively refines these divisions in a locally greedy manner, making it suitable for handling large-scale graphs.

The modularity measures the quality of the partitioning of the graph into communities. It is defined as follows:

$$\text{Modularity} = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j) \tag{5.3}$$

where:

- A_{ij} is the element in the adjacency matrix representing the connection between nodes i and j,
- k_i and k_j are the degrees of nodes i and j,
- m is the total number of edges in the graph,

- $\delta(c_i, c_j)$ is the Kronecker delta function, equal to 1 if c_i=c_j (nodes belong to the same community) and 0 otherwise.

The algorithm is shown in Algorithm 1

Algorithm 1 Louvain Community Detection Algorithm

Input Graph $G(V, E)$
Output Community assignment $C : V \to \mathbb{N}$
function LOUVAIN(G)
    Initialize each node as a separate community: $C[i] \leftarrow i$ for all $i \in V$
    $Q \leftarrow 0$
    repeat
        for each node i in V **do**
            for each neighboring community j of i **do**
                $Q_{\text{change}} \leftarrow \text{ModularityGain}(i, j, G, C)$
                if $Q_{\text{change}} > 0$ **then**
                    Move i to community j: $C[i] \leftarrow j$
                    $Q \leftarrow Q + Q_{\text{change}}$
                end if
            end for
        end for
    until No further improvement in modularity is possible **return** Community assignment C
end function
function MODULARITYGAIN(i, j, G, C)
    Compute the change in modularity by moving node i to community j
    $Q_{\text{change}} \leftarrow \text{Modularity}(G, C) - \text{Modularity}(G, C')$ **return** Change in modularity Q_{change}
end function
function MODULARITY(G, C)
    Compute the modularity of the partition C in graph G
    See the Louvain algorithm details for modularity calculation **return** Modularity value
end function

Chapter 6

Experimental Results

This project integrates a user interface (UI) with three distinct models and a chatbot, creating a comprehensive platform for efficient data interaction and analysis. The UI serves as a central hub for users to seamlessly engage with these components, providing enhanced capabilities such as predictive modeling, clustering, classification, and real-time query responses through the chatbot, Samvidh. This integration results in improved user engagement, streamlined data exploration, and enhanced decision-making within our research framework.

6.1 Models

The three models included are node classification, link prediction, and community detection. These models play crucial roles in data analysis and understanding, such as categorizing data points, predicting relationships between nodes, and identifying cohesive groups within datasets. Node classification assigns labels to data points, link prediction forecasts connections, and community detection reveals underlying structures in the data.

6.1.1 Node Classification Model

The Cora dataset, consisting of interconnected academic papers and their corresponding categories, serves as a rich resource for understanding citation relationships and topical classifications within scholarly literature. By merging the graph dataset, which encodes citation connections, with the text dataset containing paper titles and categories, a unified input is formed for analysis. Employing a Graph Convolutional Network(GCN) architecture, specifically tailored for graph-structured data, enhances our ability to discern intricate relational patterns among papers. With two convolutional layers and Rectified Linear Unit(ReLU) activation functions, the GCN efficiently captures and processes information, facilitating the extraction of nuanced insights from the interconnected web of academic citations. This model not only classifies papers based on their categories but also enables the creation of user-friendly interfaces, such as keyword-based search and category-wise filtering, empowering researchers to navigate and access relevant papers with ease and precision.

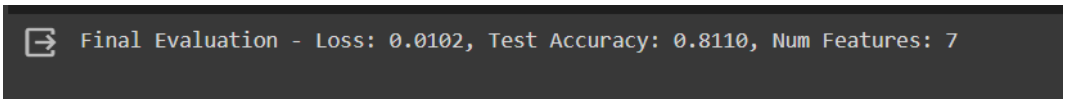


Figure 6.1: Node Classification Model Evaluation

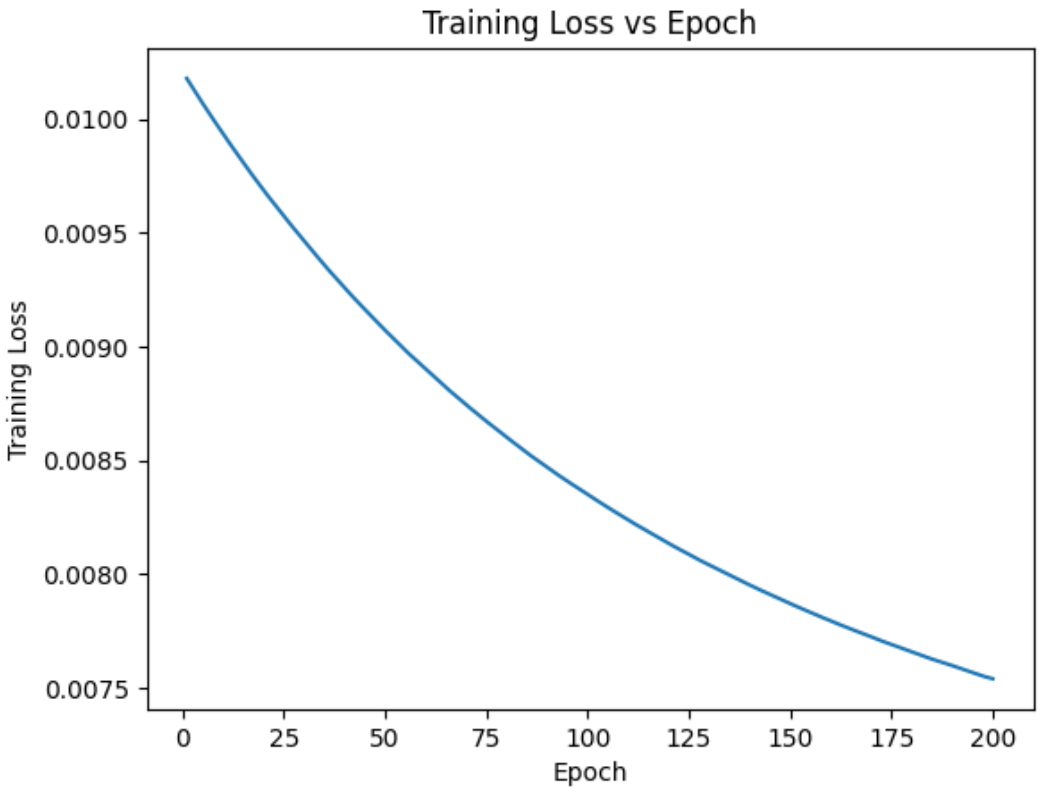


Figure 6.2: Epoch v/s Loss Graph



Figure 6.3: Keyword Search results

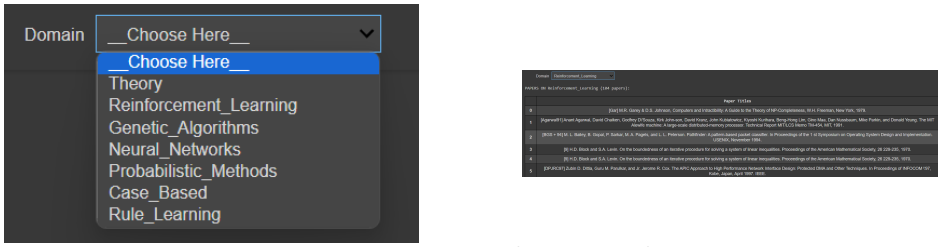


Figure 6.4: Drop down results

6.1.2 Link Prediction Model

This experiment employed a link prediction model built using a Graph Convolutional Network (GCN) on the Cora citation network dataset. The GCN

model aimed to identify missing or future links (citations) between research papers (nodes) by analyzing their relationships and features. By learning from existing connections and node characteristics, the model discovered patterns indicative of likely links. Trained with 100 epochs using cross-entropy loss and the Adam optimizer, the model comprised two GraphConv layers with ReLU activation functions in the forward pass. These GraphConv layers, specifically designed for graph convolutional operations, are initialized with parameters including the input feature size (in feats), hidden layer size (hidden size), and output feature size (out feats). In the forward method, the input graph and its features are fed through these two layers sequentially. The output of the first GraphConv layer is passed through a ReLU activation function, introducing non-linearity. The resulting activation is then passed through the second GraphConv layer, followed by another ReLU activation for further non-linearity. Finally, the model outputs the processed information. Notably, the model takes a node ID as input and predicts all the papers cited by that specific node. The model achieved an accuracy of 77 %. The Output from Link Prediction Model is shown in (Figure 6.5)

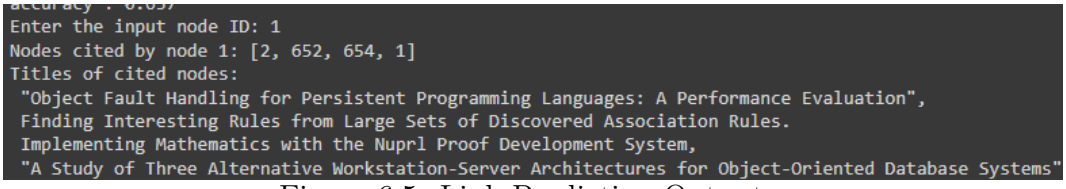


Figure 6.5: Link Prediction Output

6.1.3 Community Detection Model

```
Final Evaluation - Loss: 0.0083, Test Accuracy: 0.8110, Num Features: 7
```

Figure 6.6: evaluation of the model

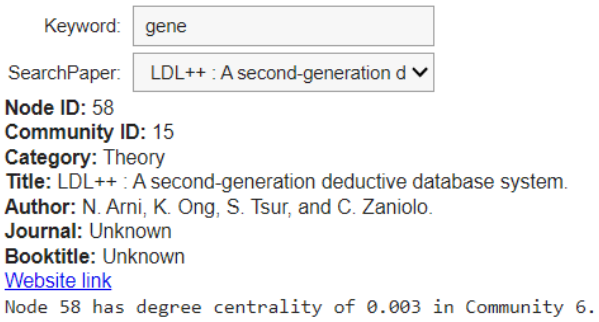
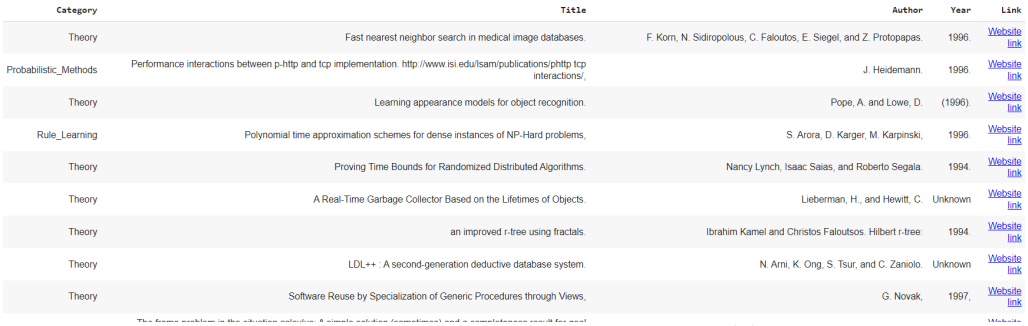


Figure 6.7: output - details and community relevance

This experiment utilized a community detection model constructed using a Graph Convolutional Network(GCN) on the Cora citation network dataset. By leveraging the graph's topology and node features, the GCN aimed to identify distinct communities or groups of related papers within the dataset. The GCN model begins with two GCNConv layers for node classification, mapping features to hidden representations and refining them for category output. ReLU activation introduces non-linearity. Through Adam optimization and backpropagation, the model minimizes cross-entropy loss, effectively categorizing nodes based on graph structure. This approach enables powerful node-level tasks, leveraging the network's topology and node attributes to identify distinct communities or groups within the dataset.The evaluation of the model is shown in (Figure 6.6). The code conducts community detection using the Louvain method on a graph constructed from edge indices. The Louvain algorithm partitions the graph into communities, assigning each node a community label. This reveals insights into cohesive communities and connectivity patterns within the graph, enhancing understanding of its underlying relationships and clusters. Notably, the model takes a paper title as input and displays the community influence by calculating its degree centrality and the list of papers in the same community.



Category	Title	Author	Year	Link
Theory	Fast nearest neighbor search in medical image databases.	F. Korn, N. Sidiropolous, C. Faloutos, E. Siegel, and Z. Protopapas.	1996.	Website link
Probabilistic_Methods	Performance interactions between p-http and tcp implementation. http://www.isi.edu/lsam/publications/phttp tcp interactions/,	J. Heidemann.	1996.	Website link
Theory	Learning appearance models for object recognition.	Pope, A. and Lowe, D.	(1996).	Website link
Rule_Learning	Polynomial time approximation schemes for dense instances of NP-Hard problems,	S. Arora, D. Karger, M. Karpinski,	1996.	Website link
Theory	Proving Time Bounds for Randomized Distributed Algorithms.	Nancy Lynch, Isaac Saias, and Roberto Segala.	1994.	Website link
Theory	A Real-Time Garbage Collector Based on the Lifetimes of Objects.	Lieberman, H., and Hewitt, C.	Unknown	Website link
Theory	an improved r-tree using fractals.	Ibrahim Kamel and Christos Faloutsos. Hilbert r-tree:	1994.	Website link
Theory	LDL++ : A second-generation deductive database system.	N. Arni, K. Ong, S. Tsur, and C. Zaniolo.	Unknown	Website link
Theory	Software Reuse by Specialization of Generic Procedures through Views,	G. Novak,	1997,	Website link

Figure 6.8: output - community table

6.2 UI

The web application user interface (UI) was meticulously crafted using Flask, HTML, CSS, and JavaScript (JS). Flask provided the framework for backend development, enabling seamless integration of data processing and rendering. HTML and CSS were utilized for structuring and styling the UI elements, ensuring a visually appealing and user-friendly layout. JavaScript added dynamic functionality, enhancing the interactivity and responsiveness of the UI components.

6.2.1 Register and Login Page

This allows the user to signup (Figure 5.2) as a new user or login (Figure 5.3) with an existing account. This provides seamless experience for both new and returning users. It empowers individuals to leverage our platform's capabilities for academic exploration and collaboration, enhancing their research endeavors within our community-oriented environment.

- **Create Account**: To access the website, begin by creating a new account.
- **Personal Information**: Provide full name, email address, and a secure password. Re-enter the password to confirm the password. This

information will be used to create account and save the data in the database.

- **Sign-up**: Entering the sign-up button will register the user to the database. After successful signing up the user will be redirected to the Login page where they can Login with email and password created while signing up.
- **Login**: If the user is a registered user, enter registered email address and password to log in. Otherwise, the user can register themselves in the sign-up page and login to the website.
- **Home Access**: Upon successful login, the user will be directed to the home page where user can access paper and visualize the papers in the community.

6.2.2 Home Page

The following outlines the key features and functionalities of our web application (Figure 5.4) for accessing research papers and analysis, highlighting its innovative tools such as 3D visualization, hover-over details, navigation bars, keyword search, and chat-bot integration.

1. **3D Visualization**:
    - Central feature showcasing dataset dynamics.
    - Interactive exploration of research papers, category, and authors.
    - Zoom functionality for detailed or broad perspectives.
2. **Hover-Over Details**:
    - Reveals essential information about research papers.

- Includes titles, authors, publication, dates, etc.
- Enhances understanding and quick access to insights.

3. **Navigation Bars**:
   - Left-side bar for seamless navigation within the application.
   - Sections include About, Logout and Home buttons
   - Enhances user experience and accessibility.
4. **Keyword driven paper-search Navigation Bar**:
   - Dedicated bar for efficient keyword-based research paper retrieval.
   - Enter any keyword or category to find relevant papers instantly.
5. **Chatbot Integration:**:
   - Intelligent chatbot,Samvidh accessible within the application.
   - Assists with queries about selected papers and additional information.
   - Natural language processing for smooth and efficient communication.

6.2.3 Keyword driven paper-search

The search (Figure 5.6) functionality within the application allows users to enter keywords or specific categories to find relevant research papers. The GNN-based node classification optimizes the data to make search process align with search queries and with categorized papers.

When a user enters a keyword or selects a category in the search box, the application utilizes the GNN model's classification results to retrieve papers

that match the user's criteria. This approach ensures that search results are tailored to the user's interests and preferences. After generating search results, users can choose any paper for further details. On selection of desired paper, users will be redirected to the Paper Details Page where users can have further details of the page.

By integrating GNN-driven dataset classification with user-initiated keyword searches and category selection, our web application delivers a sophisticated yet intuitive platform for accessing and exploring research papers. Users benefit from a refined search experience, precise result filtering, and personalized interactions, contributing to a more effective and rewarding research journey.

6.2.4 Key Paper Insights

When user select a paper from the keyword search results in our web application, they are presented with comprehensive details (Figure 5.5) about the research paper they've selected. This includes:

1. **Title**: The title of the research paper, which succinctly encapsulates its topic and scope.

2. **Authors**: The names of the authors who contributed to the research paper.

3. **Journal name**: the journal name in which the research paper was published as it is a key indicator of scholarly credibility and domain relevance for research papers.

4. **Year**: The year in which the research paper was published, a crucial indicator of the currency and historical context of research findings,

offering valuable insights into the progression and relevance of scholarly work..

5. **Abstract**: The names of the authors who contributed to the research paper, acknowledging their scholarly contributions.

6. **Citations**: A list of papers that cite the selected research paper, offering insights into its impact and influence within the academic community.

7. **Community Papers**: Papers enriched with node embeddings and weighted by GNN, when subjected to community detection on a citation graph, reveal topographic-based clusters, shedding light on thematic or structural groupings within the academic network.

8. **Accessing and Downloading Research Papers**: Users can conveniently access the paper itself by being redirected to its PDF version. This functionality allows users to read the paper directly on the platform and download it for further use, providing a seamless experience for accessing and evaluating research papers.

9. **Chat with Samvidh**: Our intelligent chatbot, Samvidh, provides an interactive platform for users to seek additional insights and ask questions about the research paper. Samvidh leverages the categorized dataset to offer personalized assistance and enhance the user experience.

This comprehensive display of information empowers users to gain a holistic understanding of the research paper's context, significance, and connections within the scholarly landscape.

6.2.5 Smart Virtual Assistant Bot(Samvidh)

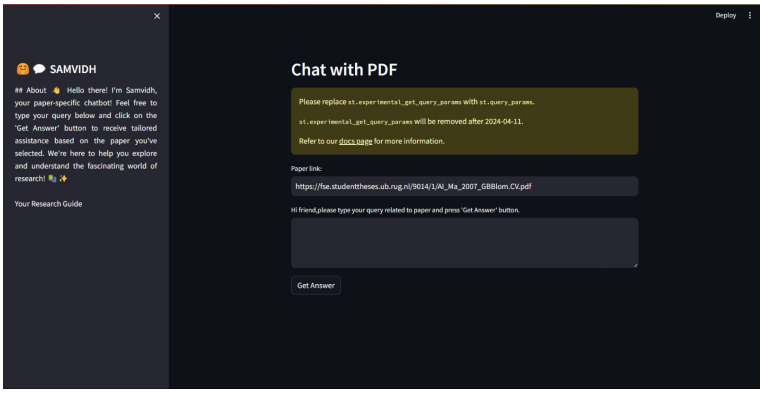


Figure 6.9: Chat with pdf

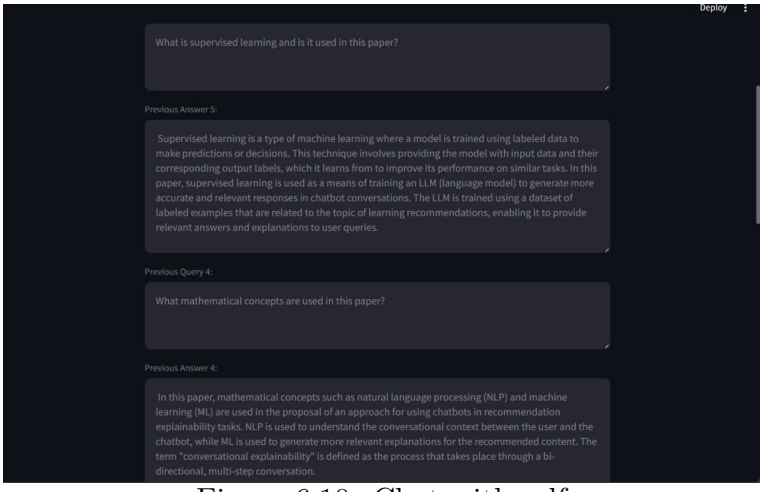


Figure 6.10: Chat with pdf

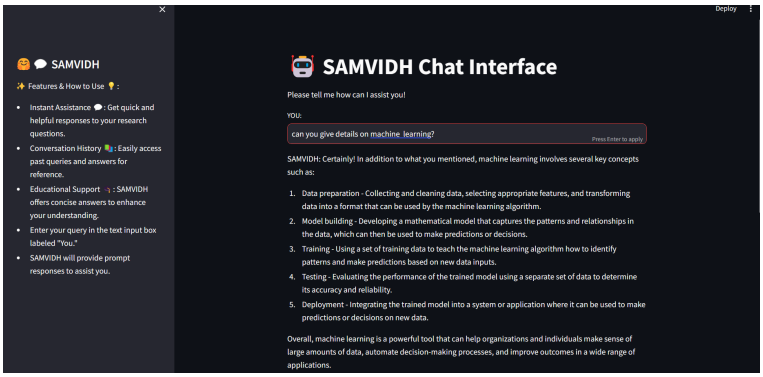


Figure 6.11: General Chatbot

The chatbot Samvidh , offers different functionalities - in two differnt parts of the website.One grants access to a vast library of academic research from

the CORA dataset, allowing users to select and explore PDFs of their interest. For those seeking deeper insights, Samvidh steps in, breaking down chosen papers into manageable sections using Word2Vec embeddings. This enables users to ask questions in natural language, leveraging Samvidh's understanding of the document's content. Supported by a Large Language Model (LLM), Samvidh provides contextually relevant responses, fostering intuitive exploration of scholarly works.

On the other hand, the second chatbot serves as a versatile companion, engaging users in general conversations and accommodating inquiries on various topics. Powered by Large Language Models, this bot adapts its responses to match the tone and complexity of each interaction. Together, these chatbots facilitate a dynamic environment where curiosity is nurtured, bridging the gap between scholarly inquiry and casual conversation, and inviting users to embark on journeys of exploration and discovery.

Chapter 7

Conclusion

Research papers, filled with scholarly content, have long served as the fundamental in accumulating knowledge across various disciplines. Our approach focused on the potential of Graph Neural Networks (GNNs) and took advantage of their ability to learn complex graph structures which would enable a more sophisticated and powerful approach to bibliometric analysis of research papers. GNNs delve deeper into the inherent structure of citation networks, enabling tasks like node classification, link prediction, and community detection which are incorporated into a user-friendly application built on a Flask framework. This system improves information retrieval functionality, personalizes paper recommendations, and provides clear visualizations of citation networks. To further enhance accessibility and interaction, a conversational chatbot named Samvidh is deployed by integrating GNN and LLM models. Samvidh excels in personalized query resolving for selected papers, leveraging the power of AI to streamline information retrieval and enhance user experience. All of the advancements sets a new standard for efficiently navigating and extracting knowledge from the vast landscape of scholarly research.

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Appendix A

Source code

A.1 Dataset

In this project, we utilized the Cora dataset[12]. We connected the Cora.planetoid and cora.cites datasets to obtain all the necessary details, including paper title, author, category, abstract, citations, publication and Year of publication.

A.2 Source Code

Github Link to the code: Click here